Apurv Anantkumar Patel

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EDUCATION

[10/2022 - Present] **MSc. Electrical Engineering – Smart Systems |** *University of Stuttgart* | Germany
Expected Grade: 1.9 (Current Grade: 2.1)

[08/2016 - 07/2020] **B. Tech Electrical Engineering – Power Electronics |** *Pandit Deendayal Energy University* | India
Final grade: 1.7
Thesis: "Single Phase Inverter with Selective Harmonics Elimination" (MATLAB/Simulink)

RESEARCH EXPERIENCE

[10/2025 – Present] **Master's Thesis** | *Fraunhofer IPA - Manufacturing Engineering and Automation*

Digital Twin-Driven Deep Reinforcement Learning (DRL) for Dexterous Pick-and-Place in Smart Manufacturing

Developing a dual-approach framework comparing end-to-end Deep Reinforcement Learning (DRL) versus hybrid DRL + Inverse Kinematics (IK) for precise assembly using a 12-joint Inspire dexterous hand with integrated tactile sensors on a UR5e manipulator arm.

Key Contributions:

- Engineered mesh-based grasp planner processing 3D CAD files to automatically generate optimal grasping points, across diverse geometries without manual annotation.
- Implemented physics-accurate digital twin in NVIDIA Isaac Lab with UR5e kinematics, Inspire hand dynamics and high-fidelity tactile sensor simulation (contact forces) for realistic grasp interactions.
- Integrated tactile sensing and force-torque feedback loops to enable compliant grasping with adaptive grip pressure based on object material properties and surface friction.
- Training DRL agents (PyTorch, RSL-RL library) with custom reward functions incorporating tactile feedback and force control, benchmarking ±2mm placement accuracy and grasp stability across object variants.
- Achieved successful task completion across multiple object geometries using hybrid policy.
- Preparing sim-to-real transfer methodology to validate simulation-trained policies.

[09/2023 – 09/2025] **Research Assistant** | *Fraunhofer IPA - Manufacturing Engineering and Automation*

Conducted comprehensive research on material behavior prediction and novel gripper design for adaptive robotic manipulation systems.

Object Behaviour Prediction:

- Investigated and implemented multiple machine learning approaches for object property estimation, including logistic regression, KNN, and CLIP models for comprehensive behavioral analysis.
- Architected and implemented a comprehensive robot-object interaction data-generation framework in NVIDIA Isaac Lab, enabling systematic collection of behavioral datasets through controlled UR10 manipulator perturbations.
- Developed data-generation framework integrating computer vision (OpenCV) for object detection and localization with structured annotation pipelines.
- Architected framework integration for graph-based ML models (AdaptiGraph framework implementation) using PyTorch, reducing prediction model setup time and enabling rapid algorithm validation for object property prediction.

Robotic Gripper Design:

- Designed and developed a 3-finger suction cup flexible robotic arm gripper capable of efficiently grasping objects with irregular shapes and orientations.

- Designed and implemented Python-based algorithms to optimize gripping points for variable triangle configurations.
- Implemented object image processing pipeline using OpenCV for morphological analysis and grasp synthesis algorithms.
- Engineered and validated robotic grippers by integrating parametric SolidWorks CAD with CoppeliaSim physics simulations, optimizing manipulation performance across diverse object geometries and orientations.

[09/2024 - 03/2025] **Research Project** | IAS **-** *Institute of Industrial Automation and Software Engineering*

Configurable Time-Series Data Generator in ROS/Gazebo with Franka Emika Panda Manipulator

Designed and developed a comprehensive software tool in ROS/Gazebo to simulate industrial tasks using the Franka Emika Panda robotic arm, aimed at generating diverse time-series datasets for safety analysis, fault detection, and risk assessment in robotic applications**.**

Key Contributions:

- Dynamic Scenario Simulation: Implemented randomization of robot speeds, motion planning algorithms, and task parameters (drilling, welding) to realistically emulate varying industrial working conditions.
- Custom End-Effector Integration: Developed new MoveIt configurations and move groups for task-specific, custom-designed end-effectors enabling precise manipulation and versatile task execution.
- Graphical Interface Development: Built an intuitive GUI using PyQt5 for seamless control over data collection, fault injection, and real-time safety violation monitoring.
- Programmable Fault Injection & Risk Assessment: Implemented Python-based mechanisms for systematic fault injection including sensor anomalies and safety violation triggering, enabling robust validation of robotic safety performance in adverse conditions.

Academic Recognition:

- Research Paper titled "*A Time-series Data Generation Tool for Risk Assessment of Robotic Applications*" Accepted at the *ESREL/SRA-E 2025 Conference.*

[09/2023 - 12/2023] **Research Assistant** | ILH - *Institute for Robust Power Semiconductor Systems*

- Designed MATLAB Simulink simulations to calculate traction load current requirements for electric vehicles for different driving cycles.
- Soldered SMD components and tested gate driver circuits for power electronics applications.

[05/2019 - 05/2020] **Research Project** | *Pandit Deendayal Energy University*

Operational Amplifier Based Low- Power dc–ac Converter

Designed and developed a pure sine-wave single-phase inverter with variable voltage and frequency control for domestic applications.

Key Contributions:

- Innovative inverter design using power operational amplifier OPA549 for variable voltage and frequency control.
- Simplified circuit architecture eliminating gate driver ICs and snubber circuits, significantly reducing design complexity while maintaining performance and reliability.
- Engineered converter for versatile domestic applications including uninterruptible power supplies (UPS) and permanent magnet brushless DC (PMBLDC) motor controllers.
- Published paper titled "*Operational Amplifier Based dc-ac Converter for Domestic Application*" at ICOEI-2020.

[02/2020 - 07/2020] **Bachelor's Thesis** | *Pandit Deendayal Energy University*

- MATLAB and Simulink Modeling: Designed and implemented detailed Simulink model for bridge-configured single-phase inverter with selective harmonic elimination (SHE).
- Fourier Analysis for Optimization: Utilized Fourier transform techniques to analyze and optimize inverter output waveform, ensuring effective minimization of targeted harmonic frequencies.
- Switching Optimization: Addressed challenges in achieving optimal switching settings to balance harmonic elimination and overall system complexity.

ADDITIONAL TECHNICAL EXPERIENCE

[06/2018 - 05/2019] **Team Lead – Electrical Design and Integration** | *Team Kaizen, PDEU*

- Played a key role in the development of an electric vehicle prototype, contributing to the advancement of electric mobility solutions.
- Developed IGBT-based power electronic drives for BLDC motors, enhancing vehicle efficiency and performance.
- Designed and implemented algorithmic code to optimize vehicle operation, ensuring smooth and efficient performance.
- Created an innovative forced air cooling system for semiconductor switches and cable harnesses, improving reliability and longevity.
- Conducted electrical wiring, troubleshooting, and in-depth analysis of controller circuits to ensure optimal functionality.

[12/2020 – 12/2021] **Commissioning and Maintenance Engineer** | *JK Cement Ltd, Nimbahera, India*

- Contributed to commissioning of various LT, HT motors and electrical equipment.
- Installation and configuration of smart load center equipped with Schneider's LTMR for LT motors optimization.
- Developed and implemented preventive maintenance schedules for equipment and machinery.
- Collaborated with cross-functional teams to ensure equipment operated safely and efficiently.

PROFESSIONAL DEVELOPMENT

- Completed a hands-on Design Factory Stuttgart semester course in innovation and design thinking at the ARENA2036 research campus, solving real-world challenges in interdisciplinary teams using agile methods (Design Thinking, Scrum).
- Built strong skills in user empathy, creative problem-solving, iterative prototyping, and systems thinking to drive responsible, experimentation-driven innovation with industry partners.
- Served as a technical club member and trainer for Cretus, the Robotics and Automation Club of PDEU.

TECHNICAL SKILLS & PROFESSIONAL ABILITIES

- **Programming:** Python: Advanced | C/C++: Intermediate | MATLAB/Simulink: Advanced | Git: Advanced
- **Robotics & Simulation:** ROS/ROS2: Advanced | Isaac Sim/Lab: Advanced | Gazebo: Advanced | MoveIt: Advanced | CoppeliaSim: Intermediate
- **Design & CAD:** SolidWorks: Advanced | Altium Designer: Intermediate
- **Electronics Tools:** Vector CANoe: Basic | LTspice: Advanced | Multisim: Advanced

LANGUAGE SKILLS

- **English:** Full Professional Proficiency (IELTS 7.5 Bands / C1)
- **German:** Limited Working Proficiency (A1 & Learning)
- **Hindi, Gujarati:** Native / Bilingual Proficiency

HOBBIES

- **Playing Badminton, Table Tennis, and Volleyball:** Two-time Regional Table Tennis representative (U17 & U19)
- **Reading Books**

PUBLICATIONS & CONFERENCE PRESENTATIONS

- Y. Ma, **A. Patel**, D. Kurian, J. Siebert, S. Vock and A. Morozov, "A Time-series Data Generation Tool for Risk Assessment of Robotic Applications," 35th European Safety and Reliability Conference and 33rd Society for Risk Analysis Europe Conference (ESREL–SRA-E), Singapore, 2025.
- **A. Patel** and A. V. Sant, "Operational Amplifier Based Low Power dc-ac Converter for Domestic Applications," *2020 4th International Conference on Trends in Electronics and Informatics (ICOEI)(48184)*, Tirunelveli, India, 2020, pp. 58-63, doi: 10.1109/ICOEI48184.2020.9142939.